RECLAIM can an agent reproduce a paper's central claim from what its authors released?

1 THE TASK



CQ-DINO: Mitigating Gradient Dilution via Category Queries…

NeurIPS 2025 · arXiv 2503.18430

Match target

cheapest config that proves the paper's claim

```
metric  AP / AP50 / AP75
scope   V3Det val · 29,821 imgs
target  46.3 / 51.5 / 48.4 ±5%
```

configuration withheld

What the authors released

✓ data ✓ code ✓ weights

AVAILABILITY TIER

- ▸ **Run** execute what shipped
- Retrain weights withheld
- Reimplement code withheld too

the availability cliff

success falls as the release thins

budget 8 H100-h

×100 papers

2 THE RUN

Reproduction agent deepseek-v4

workspace_bash · write_file · fetch_url · run_gpu

run 2889575 · 107 rounds · 6.0 of 8 H100-h

```
PLAN · r0-4
● [plan] AP 46.3 on V3Det · design eval, run it
FETCH · r5-8
○ $ git clone github.com/…/CQ-DINO
○ $ gdown …swinb1k_v3det.pth # 1.89 GB
BUILD · r9-61
○ $ pip install mmcv==2.2.0
●   mmcv/_ext…so: undefined symbol
○ wheel built for an older torch. build it
○ $ FORCE_CUDA=1 pip install -e mmcv/
RUN · r62-64
● $ run_gpu tools/test.py # 29,821 imgs        GPU
●   srun: task 0: Out Of Memory
RESCUE & SCORE · r65-92
○ OOM again, but the preds survived in /tmp
○ $ cp /tmp/results.bbox.json evidence/
● $ run_gpu chunked_eval.py # score them       GPU
VERIFY & REPORT · r93-106
○ my scorer agrees with COCOeval, twice
○ $ write_file report.json
```

bbox_mAP 0.463 0.515 0.484 = paper's 46.3 / 51.5 / 48.4 ✓

● plan ○ *thought* ○ step ● GPU run ● error $ agent command

3 THE REPORT

report.json claims: "reproduced"

evidence/ eval.log · commands.log

results.bbox.json · patches/

every number traces to a logged run

report + evidence

4 THE AUDIT

Adversarial audit

re-derives the verdict from the evidence

- ✓ metric recomputed from raw predictions
- ✓ exact checkpoint, sibling rows ruled out
- ✓ $AP_{S/M/L}$ values not in the paper: no echo
- ⚑ training arm untested (released ckpt only)

THE RUN CLAIMS	vs	THE AUDIT FINDS
AP 46.3 ✓		AP 46.3 ✓

RECLAIM scores the audited verdict